

OpenConstructionEstimate — South Africa / Johannesburg

****Construction Work Items, Components & Resources****

Property	Value
Reference region	South Africa / Johannesburg
ISO country	ZA
Language	`en`
Currency	`ZAR` (ISO 4217)
Source track	`UK\_GBP` (United Kingdom / London)
Generated	2026-04-27
Work items	900,225 rows
Unique resources	6,670

Available Formats

Format	File
Parquet	`ZA\_JOHANNESBURG\_workitems\_costs\_resources\_DDC\_CWICR.parquet`
Excel (formatted)	
`ZA\_JOHANNESBURG\_workitems\_costs\_resources\_DDC\_CWICR\_FORMATTED.xlsx`	
Excel (simple)	`ZA\_JOHANNESBURG\_workitems\_costs\_resources\_DDC\_CWICR\_SIMPLE.xlsx`
Catalog CSV	`DDC\_CWICR\_ZA\_JOHANNESBURG\_Catalog.csv`
Catalog XLSX	`DDC\_CWICR\_ZA\_JOHANNESBURG\_Catalog.xlsx`
Qdrant snapshot	
`ZA\_JOHANNESBURG\_workitems\_costs\_resources\_EMBEDDINGS\_3072\_DDC\_CWICR.snapshot` |

How this track was built

This track was generated from `UK\_GBP` by
`0\_Workflow and Pipelines CWICR/python/11-country-track-builder/add\_country\_track.py`.

- ****Norms**** (labour hours, machine hours, resource quantities) are identical to the source track — Resource-Based Costing methodology treats norms as country-agnostic physical first principles.
- ****Prices**** are derived via the cascade
`type\_factors` → `location\_factor` → optional national overrides`. Type factors come from OECD wage indexes (labour), construction PPP (material), and ECB FX (equipment). FX snapshot date is recorded in the per-resource columns.
- ****Language**** columns are translated where target language differs from source. Existing tracks are used as parallel-text seeds.

Data Structure

85+ columns organised into:

- Classification hierarchy (10 cols)

- Rate / work-item identifiers (11 cols)
- Resource decomposition (7 cols)
- Labour metrics (11 cols)
- Machinery & equipment (12 cols)
- Price aggregates (16 cols)
- Mass / service / regional markers (~18 cols)

See the repository-level [DATA_DICTIONARY.md](../DATA_DICTIONARY.md) for the complete column-by-column reference.

Qdrant collection

Vector index built from concatenated localised text fields, encoded with OpenAI `text-embedding-3-large` (3072-dim, cosine).

```
qdrant-client snapshot upload \
--collection ddc_za_johannesburg \
--snapshot ZA_JOHANNESBURG_workitems_costs_resources_EMBEDDINGS_3072_DDC_CWICR.snapshot
```

Licence

Same as the parent dataset: CC BY 4.0 for data, see [LICENSE-DATA.txt](../LICENSE-DATA.txt). Code: see [LICENSE-CODE.txt](../LICENSE-CODE.txt).